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Systems and methods for mobile digital currency future exchange

Abstract

A system for a mobile digital currency exchange is disclosed. The system may receive a sign-in data. The system may generate an authentication request based on the sign-in data. The system may receive an authentication data. The system may display a trading interface comprising a portfolio drawer based on the authentication data, wherein the portfolio drawer is configured to display an instrument detail and an account summary.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a non-provisional of, and claims priority to and benefit of, U.S. Provisional Application No. 62/895,573 filed on Sep. 4, 2019 and entitled “SYSTEMS AND METHODS FOR MOBILE DIGITAL CURRENCY FUTURE EXCHANGE.” This application is also a non-provisional of, and claims priority to and the benefit of, U.S. Provisional Application No. 63/054,172 filed on Jul. 20, 2020 and entitled “SYSTEMS AND METHODS FOR MOBILE DIGITAL CURRENCY FUTURE EXCHANGE.”

Each of the foregoing applications is hereby incorporated by reference in its entirety for all purposes (except for any subject matter disclaimers or disavowals, and except for any conflict with the present disclosure, in which case the express disclosure herein shall control).

TECHNICAL FIELD

(1) The present disclosure generally relates to mobile devices, and in particular to systems and methods for mobile and automated command and control of exchange trading platforms and applications.

BACKGROUND

(2) Traditionally, exchange and electronic trading platforms are connected via private networks and dedicated terminals. Such platforms tend to be limited in capability and may only be accessible during market hours or may otherwise be restricted in geographic or temporal availability. Additionally, options for viewing positions and executing orders may be limited. Accordingly, improved mobile trading platforms, for example platforms for trading cryptocurrency derivatives, remain desirable.

SUMMARY

(3) In various embodiments, systems, methods, and articles of manufacture (collectively, the "system") for mobile digital currency exchanges are disclosed. In various embodiments, the system may receive a sign-in data. The system may generate an authentication request based on the sign-in data. The system may receive an authentication data. The system may display a trading interface comprising a portfolio drawer based on the authentication data, wherein the portfolio drawer is configured to display an instrument detail and an account summary.

(4) In various embodiments, may receive an instrument details selection. The system may start an order flow process in response to the instrument details selection. The system may receive an order flow input. The system may generate a platform command set based on the order flow input. The system may control a trading platform based on the platform command set. In various embodiments, the order flow input comprises at least one of a buy order, a sell order, an order modification, a leverage setting, or a close position command. In various embodiments, the system may start an anonymous mode process in response to the sign-in data. In various embodiments, the system may determine a user identity based on the authentication data, wherein the authentication data includes biometric data.

(5) The foregoing features and elements may be combined in various combinations without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings. The contents of this section are intended as a simplified introduction to the disclosure, and are not intended to limit the scope of any claim.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may be obtained by referring to the detailed description and claims when considered in connection with the drawing figures, wherein like numerals denote like elements.

(2) FIG. **1** is a block diagram illustrating various system components of a mobile digital currency exchange system, in accordance with various embodiments;

(3) FIG. **2** illustrates a launch interface of a mobile digital currency exchange system, in accordance with various embodiments;

(4) FIG. **3** illustrates a currency pair display of a mobile digital currency exchange system, in accordance with various embodiments;

(5) FIG. **4** illustrates a registration interface of a mobile digital currency exchange system, in accordance with various embodiments;
(6) FIG. **5**A illustrates a sign-in interface of a mobile digital currency exchange system, in accordance with various embodiments;
(7) FIG. **5**B illustrates an authentication interface of a mobile digital currency exchange system, in accordance with various embodiments;
(8) FIG. **5**C illustrates a push notification prompt of a mobile digital currency exchange system, in accordance with various embodiments;
(9) FIG. **6** illustrates a market/portfolio interface of a mobile digital currency exchange system, in accordance with various embodiments;
(10) FIG. **7**A illustrates an account configuration interface of a mobile digital currency exchange system, in accordance with various embodiments;
(11) FIG. **7**B illustrates a notification settings interface of a mobile digital currency exchange system, in accordance with various embodiments;
(12) FIG. **7**C illustrates a security settings interface of a mobile digital currency exchange system, in accordance with various embodiments;
(13) FIG. **7**D illustrates a user guide page of a mobile digital currency exchange system, in accordance with various embodiments;
(14) FIG. **7**E illustrates a help request page of a mobile digital currency exchange system, in accordance with various embodiments;
(15) FIG. **8**A illustrates an account balance page of a mobile digital currency exchange system, in accordance with various embodiments;
(16) FIG. **8**B illustrates a position details page of a mobile digital currency exchange system, in accordance with various embodiments;
(17) FIG. **8**C illustrates a position details page of a mobile digital currency exchange system, in accordance with various embodiments;
(18) FIG. **8**D illustrates an amend position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(19) FIG. **9**A illustrates a market data page of a mobile digital currency exchange system, in accordance with various embodiments;
(20) FIG. **9**B illustrates a market data page of a mobile digital currency exchange system, in accordance with various embodiments;
(21) FIG. **9**C illustrates an order book page of a mobile digital currency exchange system, in accordance with various embodiments;
(22) FIG. **10**A illustrates a leverage interface of a mobile digital currency exchange system, in accordance with various embodiments;
(23) FIG. **10**B illustrates a leverage interface of a mobile digital currency exchange system, in accordance with various embodiments;
(24) FIG. **11**A illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;
(25) FIG. **11**B illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;
(26) FIG. **11**C illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;
(27) FIG. **11**D illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;
(28) FIG. **11**E illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;
(29) FIG. **11**F illustrates an order interface of a mobile digital currency exchange system, in accordance with various embodiments;

(30) FIG. **12**A illustrates a close position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(31) FIG. **12**B illustrates a close position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(32) FIG. **13**A illustrates an order reporting display of a mobile digital currency exchange system, in accordance with various embodiments;
(33) FIG. **13**B illustrates an order reporting display of a mobile digital currency exchange system, in accordance with various embodiments;
(34) FIG. **13**C illustrates an order reporting display of a mobile digital currency exchange system, in accordance with various embodiments;
(35) FIG. **14**A illustrates a portfolio view page of a mobile digital currency exchange system, in accordance with various embodiments;
(36) FIG. **14**B illustrates a portfolio details view page of a mobile digital currency exchange system, in accordance with various embodiments;
(37) FIG. **15**A illustrates an advanced long position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(38) FIG. **15**B illustrates an advanced long position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(39) FIG. **15**C illustrates editing a long position in the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;
(40) FIG. **15**D illustrates editing a long position in the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;
(41) FIG. **16**A illustrates an advanced short position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(42) FIG. **16**B illustrates an advanced short position interface of a mobile digital currency exchange system, in accordance with various embodiments;
(43) FIG. **16**C illustrates editing a short position in the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(44) FIG. **16**D illustrates editing a short position in the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(45) FIG. **17** illustrates details of a dynamic margin position indicator in a mobile digital currency exchange system, in accordance with various embodiments;
(46) FIG. **18**A illustrates operation of a dynamic margin position indicator in editing a short position of the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(47) FIG. **18**B illustrates operation of a dynamic margin position indicator in reducing a short position of the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(48) FIG. **18**C illustrates operation of a dynamic margin position indicator in closing a short position of the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(49) FIG. **18**D illustrates operation of a dynamic margin position indicator in reversing a short position of the advanced short interface of a mobile digital currency exchange system, in accordance with various embodiments;
(50) FIG. **19**A illustrates operation of a dynamic margin position indicator in editing a long position of the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;
(51) FIG. **19**B illustrates operation of a dynamic margin position indicator in reducing a long position of the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;

(52) FIG. **19**C illustrates operation of a dynamic margin position indicator in closing a long position of the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;
(53) FIG. **19**D illustrates operation of a dynamic margin position indicator in reversing a long position of the advanced long interface of a mobile digital currency exchange system, in accordance with various embodiments;
(54) FIG. **20** illustrates a quantity input interface of a mobile digital currency exchange system, in accordance with various embodiments;
(55) FIG. **21** illustrates a first launch process of a mobile digital currency exchange system, in accordance with various embodiments;
(56) FIG. **22** illustrates a market view process of a mobile digital currency exchange system, in accordance with various embodiments;
(57) FIG. **23** illustrates a set leverage process of a mobile digital currency exchange system, in accordance with various embodiments;
(58) FIG. **24** illustrates a second launch process of a mobile digital currency exchange system, in accordance with various embodiments;
(59) FIG. **25** illustrates an instrument details process of a mobile digital currency exchange system, in accordance with various embodiments;
(60) FIG. **26** illustrates a close position process of a mobile digital currency exchange system, in accordance with various embodiments;
(61) FIG. **27** illustrates a third launch process of a mobile digital currency exchange system, in accordance with various embodiments;
(62) FIG. **28** illustrates a portfolio drawer process of a mobile digital currency exchange system, in accordance with various embodiments;
(63) FIG. **29**A illustrates an order modification process of a mobile digital currency exchange system, in accordance with various embodiments;
(64) FIG. **29**B illustrates an order modification process of a mobile digital currency exchange system, in accordance with various embodiments;
(65) FIG. **30**A illustrates a maintenance process of a mobile digital currency exchange system, in accordance with various embodiments;
(66) FIG. **30**B illustrates a maintenance process of a mobile digital currency exchange system, in accordance with various embodiments;
(67) FIG. **31**A illustrates a sign-in process of a mobile digital currency exchange system, in accordance with various embodiments;
(68) FIG. **31**B illustrates a sign-in process of a mobile digital currency exchange system, in accordance with various embodiments;
(69) FIG. **32**A illustrates a sign-in process of a mobile digital currency exchange system, in accordance with various embodiments; and
(70) FIG. **32**B illustrates a biometric authentication process of a mobile digital currency exchange system, in accordance with various embodiments.

DETAILED DESCRIPTION

(71) The following description is of various exemplary embodiments only, and is not intended to limit the scope, applicability or configuration of the present disclosure in any way. Rather, the following description is intended to provide a convenient illustration for implementing various embodiments including the best mode. As will become apparent, various changes may be made in the function and arrangement of the elements described in these embodiments without departing from the scope of the appended claims.
(72) For the sake of brevity, conventional techniques for mobile device application design and implementation, as well as conventional mobile device communications techniques, interface elements, and so forth, and/or the like, may not be described in detail herein. Furthermore, the

connecting lines shown in various figures contained herein are intended to represent exemplary functional relationships and/or physical or communicative couplings between various elements. It should be noted that many alternative or additional functional relationships or communicative connections may be present in a practical system or related methods of use, for example a mobile trading application for cryptocurrency derivatives.

(73) Various shortcomings of mobile device applications can be addressed by utilizing mobile device applications and/or related cloud-based systems configured in accordance with principles of the present disclosure. For example, the present system improves upon existing technology by optimizing displays into modes which carry out platform activities including setting leverage, by providing a single assembly flow for any order type, monitoring each of a balance, an order set, and positions associated with an account, and by visualizing risk levels associated with a specific position.

(74) In various exemplary embodiments, the system may provide a greater level of sophistication and/or control for digital currency exchange systems. For example, data may be gathered from multiple data sources comprising multiple dissimilar rows and columns and may be distributed across multiple platforms. While prior art systems typically include the technical problem of limited availability, multiple user interactions to execute an order or view displays, obfuscating position specific risk and/or the like, the current system provides a technical solution by tending to enable a single flow irrespective or order type and by visualizing position specific risk levels. In this regard, the system may enable accelerated trading via an integrated order flow and optimized data presentation. As such, the system may eliminate or reduce information gaps, reduce re-entry of data, and reduce record duplication, and reduce development time. The system may also reduce the cost of development or system processing time for data entry, reduce network utilization, and/or reduce data storage overhead. The system may increase data reliability and/or accuracy by enabling comparison of data between environments at an increased frequency. The system may also reduce redundant or duplicate comparison tasks, thereby reducing a demand for system resources. The system may simplify data acquisition and enhance the user experience by decreasing the number of user interactions (e.g., for an order within the order flow, all options to assemble all supported order types are presented to quickly view full details and modify only the options necessary therefore the user does not need to switch order types and reset previous selections). Moreover, benefits of the present disclosure may apply to any suitable electronic exchange or trading platform.

(75) Processes disclosed herein improve the functioning of the computer. In various s embodiments, order processing speeds may be accelerated via a reduced interaction set enabled by a single inline order flow. The system may further reduce inputs by enabling a universally accessible portfolio drawer thereby connecting order functionality with the context of a data display. In various embodiments, the system may reduce graphics processing overhead by tending to reduce the need for multiple scrolls and clicks to access the data that is most frequently viewed. For example, the system may display a standardized data set including balances and market positions on sign-in. Similarly, the process increases the reliability and speed of data presentation by enabling direct comparison of real time data between environments on the basis of metadata elements. The system may automatically update its context to help provide information about a current position size and contract specific uPnL. The system increases the reliability and speed of analysis by enabling a real-time view of positions and margin levels across asset classes. For example, the system may enable position modification actions in a single click with buy, sell, close position, close order, and close stops shown in context of a contract display. Thus, the system may tend to enable context based prediction of user inputs and further accelerate order flows. In this regard, by transmitting, storing, and/or accessing data using the processes described herein, the informational utility of the data is improved, and errors are reduced. Such improvements may also increase the efficiency of the network by reducing a portion of duplicated effort as additional data sources are identified for comparison.

(76) In various embodiments, and with reference now to FIG. **1**, a system **100** may comprise an application server **102**, a user device **104**, and a trading platform **200**. Any of these components may be outsourced and/or be in communication with the data comparator application server **102** and/or trading platform **200** via a network such as, for example a first network **106** and a second network **108**.

(77) System **100** may be computer-based, and may comprise a processor, a tangible non-transitory computer-readable memory, and/or a network interface, along with other suitable system software and hardware components. Instructions stored on the tangible non-transitory memory may allow system **100** to perform various functions, as described herein. In various embodiments, the application server **102** and/or trading platform **202** may be configured as a central network element or hub to access various systems, engines, and components of system **100**. The application server **102** may comprise a network (e.g., network **106**), a computer-based system, and/or software components configured to provide an access point to various systems, engines, and components of system **100**. The application server **102** may be in operative and/or electronic communication with user devices **104** via the first network **106** and the trading platform **200** via the second network **108**. In this regard, the application server **102** may allow communication from the user devices **104** to systems, engines, and components of system **100** (such as, for example, trading platform **202**). In various embodiments, the application server **102** may receive commands and/or metadata from the user devices **104** and may pass replies to the user devices **104**.

(78) In various embodiments, application server **102** may include one or more computing devices described above, rack mounted servers, and/or virtual machines providing load balancing, application services, web services, data query services, data transfer services, reverse proxy services, or otherwise facilitating the delivery and receipt of data across networks (**106**, **108**).

(79) In various embodiments, a user device **104** may comprise software and/or hardware in communication with the system **100** via a network (e.g. network **106**) comprising hardware and/or software configured to allow a user, and/or the like, access to the application server **102**. The user device may comprise any suitable device that is configured to allow a user to communicate with a network and the system **100**. The user device may include, for example, a personal computer, personal digital assistant, cellular phone, kiosk, and/or the like and may allow a user to transmit comparison requests to the system **100**. In various embodiments, the user device **104** described herein may run a web application or native application to communicate with application server **102**. A native application **110** may be installed on the user device **104** via download, physical media, or an app store, for example. The native application **110** may utilize the development code base provided for use with the operating system and capable of performing system calls to manipulate the stored and displayed data on the user device **104** and communicates with application server **102**. A web application may be web browser compatible and written specifically to run on a web browser. The web application may thus be a browser-based application that operates in conjunction with application server **102**.

(80) In various embodiments, the native application **110** running on the user device **104** may be in communication with the application server **102** to support real-time updates. For example, data pertaining to the trading platform **200** may synchronize across the various user devices **104** used by any number of users interacting with the application server **102** and/or trading platform **200**. In this regard, the application server **102** may serve data from trading platform **200** to each of the user devices **104** and may serve commands from the user devices **104** to the trading platform **200**. In various embodiments, application server **102** may apply access permissions to restrict the data transmitted between the networks (**106**, **108**) and/or the various components of system **100**. Users may be authenticated on the native application **110**, for example, via a user name, password, dual factor authentication, private cryptographic key, one-time password, security question, biometrics, or other suitable authentication techniques know to those skilled in the art.

(81) In various embodiments a trading platform **200** in a digital currency future exchange system

(e.g., system **100**) is disclosed. Trading platform **200** may include an API interface, an order book, an order processing engine, a leverage engine, a matching engine, and/or the like.

(82) With additional reference to FIG. **2**, a launch interface **202** of platform **200** is illustrated in accordance with various embodiments. System **100** may display the launch interface **202** in response to executing via the user device **104** the native application **110**. In various embodiments, the system may determine a connection status **204** between the trading platform **200** and the user device **104** and display a connection status indicator via the launch interface **202**. The system may display one or more details of market data **206** such as, for example, a currency price, a currency pair, a derivatives price, an index price, and/or the like. The launch interface **202** may be configured to receive one or more inputs from the user device **104** such as a register action and a sign-in action. For example, the system may receive the register action from the user device **104** in response to sensing a user interaction with the register button **208** and may receive the sign-in action from the user device **104** in response to a user interaction with the sign-in button **210**. In various embodiments, launch interface **202** may be configured to enable interactions with the elements of market data **206**. In response to determining an interaction with an element of the market data, the system may generate a market detail request associated with the element.

(83) For example, the system may register an interaction with the “Bitcoin (XBT)” market data **206** and, in response, may generate (e.g., via the native application) a Bitcoin (XBT)” market data detail request which may be received by the trading platform **200**. In response to receiving the market data detail request, the system may display expanded market details associated with the market detail request. For example, the system may display a currency pair display as illustrated in FIG. **3**. With additional reference to FIG. **3**, currency pair display **300** may be associated with the “Bitcoin (XBT)” market data and displayed in response to receiving the associated market detail request. The currency pair display **300** may include various market details associated with the currency pair such as, for example, a current price **302**, an index price **304**, a 24 hour % change in price **306**, an exchange volume **308**, an open instrument value **310**, and/or the like. In various embodiments, the currency pair display **300** may include a real-time graphical display of the market (i.e., market graph **312**). In various embodiments, the market graph **312** may be a candlestick chart and may be selectable across various time intervals. Currency pair display **300** may be configured to receive market graph time interval inputs via radio buttons **314**. In various embodiments, the currency pair display **300** may include order book data **316**.

(84) In various embodiments and with additional reference to FIG. **4** a registration interface **400** of system **100** is illustrated in accordance with various embodiments. The system may display the registration interface **400** in response to the register action. The registration interface **400** may be configured to receive one or more elements of registration data which may be associated by the system with a user profile and/or the like. In various embodiments the system may be configured to receive an email address **402**, a password **404**, a country or region **405**, a first name **480**, a last name **410**, biometric information (e.g., fingerprint, faceprint, voice sample, iris image, etc.), device fingerprint (e.g., metadata associated with the user device **104** such as, for example, mac address, operating system data, network information, version information, etc.), and/or the like. In response to receiving the registration data the system may generate a user account and associate the registration data therewith as account information. In various embodiments, the system may generate an authentication token and associate the authentication token with the user account. In this regard, the system may support token authentication such as two-factor authentication and/or the like.

(85) In various embodiments and with additional reference to FIG. **5**A, a sign-in interface **500** of system **100** is illustrated. The sign-in interface **500** may be configured to receive sign-in data. For example, the system may receive an email address **502**, a password **504**, and a two-factor token **506** via the sign-in interface **500**. In various embodiments, the system may generate a sign-in request via the user device **104** in response to an interaction with the sign in button **508**. The sign in request

may comprise the sign-in data which may be compared to the account information as part of an authentication process. In various embodiments, the system may generate an authentication request based on the sign-in data.

(86) In various embodiments and with additional reference to FIG. 5B, an authentication interface **510** of system **100** is illustrated. The system may display the authentication interface **510** in response to the authentication request. The authentication interface **510** may include a prompt to set or enter authentication data such as, for example, a numeric passcode, a touch pattern, a biometric data, and/or the like. In various embodiments, authentication interface **510** may be configured to receive the authentication data which may be associated with the user data as account information. For example, the authentication interface **510** may be configured to receive a numeric passcode via numeric keypad **512**. The system may receive the authentication data with the sign-in data and compare the authentication data and the sign-in data with the account information to enable access to the various systems, features, and engines of the trading platform **200**. With brief additional reference to FIG. 5C, the system may display a push notification prompt **514** in response to receiving the authentication data. The system may enable push notifications from the trading platform **200** to the user device **104** in response to receiving an ‘allow’ interaction with the push notification prompt **514**.

(87) In various embodiments and in response to receiving the authentication data and the sign-in data, the system may display a splash screen and/or a market/portfolio interface **600** as illustrated in FIG. **6**. The market/portfolio interface **600** may include selectable market data for cryptocurrencies **602** and/or currency pairs **604**. The system may display selectable market data for one or more instruments associated with the cryptocurrency **602** and/or currency pairs **604**. For example, the system may display futures contracts **606** associated with the cryptocurrency **602** such as a perpetual contract, or various finite contracts **608** (e.g., closing at one month intervals ‘September 27’, ‘December 27’, and/or the like). The system may display current price data **610**, index price data, strike price data, volume data, velocity data, or any other market data associated with the cryptocurrencies **602**, currency pairs **604**, and or instruments. In various embodiments, the market/portfolio interface **600** may be configured to display an account balance drawer **612**. In various embodiments, the market/portfolio interface **600** may be configured to display an account settings icon **614**. In response to receiving an interaction with the account settings icon **614**, the system may display an account configuration interface.

(88) With additional reference to FIG. 7A, an account configuration interface **700** of system **100** is illustrated in accordance with various embodiments. The account configuration interface **700** may display various elements of account information associated with the user account such as a user name **702** and an email **704**. In various embodiments, the account configuration interface **700** may be configured to enable access to one or more configuration settings pages of the system such as, for example, via interaction with a notifications settings button **706**, a security settings button **708**, a user guide button **710**, an a help request button **712**. In various embodiments, the account configuration interface **700** may be configured to display informational pages in response to receiving an interaction via a terms of service button **714** and a privacy notice button **716**. In various embodiments, the system may be configured to terminate an authenticated session in response to receiving an interaction with a sign-out button **718**. In this regard, the system may disable access to the various systems, features, and engines of the trading platform **200** in response to the interaction with the sign-out button **718**. In various embodiments, the account configuration interface **700** may be configured to display the account balance drawer **612**.

(89) In various embodiments and with additional reference to FIG. 7B, in response to receiving the interaction with the notifications settings button **706** the system may display a notification settings interface **720**. The notification settings interface **720** may be configured to receive notifications settings inputs and, in response, enable or disable various push notifications from the trading platform **200** to the user device **104**. For example, notification settings interface **720** may be

configured to receive interactions with one or more switches to enable/disable notifications for deposit pending **722**, deposit confirmation **724**, automatic delivering events **726**, liquidation events **728**, withdrawal confirmation **730**, and alternate device sign-in **732**. For example, the system may compare device fingerprints associated with a user account as account information. The system may generate an alternate device sign-in notification where a device fingerprint received with the sign-in data does not match the device fingerprint stored as the account information. Stated another way, the system may receive a first device fingerprint and a first sign-in data from a first user device and a second device fingerprint and the first sign in-data from a second user device. The system may generate the alternate device sign-in notification in response to receiving the second device fingerprint and the first sign in-data from the second user device. The system may transmit the alternative device sign-in notification to the first user device. In various embodiments, the notification settings interface **720** may be configured to display the account balance drawer **612**.

(90) In various embodiments and with additional reference to FIG. 7C, in response to receiving the interaction with the security settings button **708** the system may display a security settings interface **734**. The security settings interface **734** may be configured to enable and/or disable the authentication interface **510** and associated system processes. For example, the authentication interface **510** may be disabled in response to receiving receive interactions with an enable/disable command switch **736**. In various embodiments, the security settings interface **734** may be configured to edit or reconfigure the authentication data. For example, the system may enable editing of the numeric passcode via an interaction with the edit passcode button **738**. In various embodiments, the security settings interface **734** may be configured to display the account balance drawer **612**.

(91) In various embodiments and with additional reference to FIG. 7D, in response to receiving the interaction with the user guide button **710** the system may display a user guide page **740**. In various embodiments, the system may start a web punch out process and the native app **110** may communicate with one or more web servers to display the a user guide page **740**.

(92) In various embodiments and with additional reference to FIG. 7E, in response to receiving the interaction with the help request button **712**, the system may display a help request page **742**. In various embodiments, the system may start a web punch out process and the native app **110** may communicate with one or more web servers to display the a help request page **742**. In various embodiments, the help request page **742** may be configured to receive help request information. The help request information may comprise text inputs such as, for example, an email address **744**, a subject line **746**, a problem description **748**, and/or the like. In various embodiments, the help request information may include graphical data or data files which may be included as an attachment **750**.

(93) In various embodiments and in response to receiving an interaction with the account balance drawer **612**, the system may display an account balance page **800**. For example, the system may receive and interaction from the user device **104** such as dragging the account balance drawer **612** upward and, in response, expand the account balance drawer **612** to display the account balance page **800**. In various embodiments, the account balance page **800** may be configured to display account balance and position data an associated with a user account. The account balance page may display a total available balance **802**, a total margin balance **806**, a total margin composition indicator **808**, an overall PnL **804** (which may be realized or unrealized) and/or the like. The total margin composition indicator **808** may be a graphical indicator such as, for example, a bar graph showing the total values of all positions and all current orders as a portion of total available margin. Each component may be displayed in different colors and thereby serve to quickly distinguish between the all positions value, the open orders value, and the total available margin value. For example, the bar graph may fill toward the right of the account balance page **800** as margin is used on various open positions and orders and thereby reduce the relative size of the displayed total available margin in the bar graph.

(94) In various embodiments, the account balance page **800** may include a position breakout by currency pair providing details of the components of the overall position associated with the user account such as positions categorized in terms of currency pairs. For example, the system may display a XBTUSD position details **810** and a ETHUSD position details **812**. Either of the position details displays (**810**, **812**) may include associated market data such as a count of active orders and/or order types **814**, position size **816**, return calculation **818**, PnL calculations **820**, position value **822**, and a liquidation details frame **824**.
(95) In various embodiments and with additional reference to FIGS. **8**B and **8**C, in response to receiving an interaction with one of the position details (**810**, **812**) the system may display a position details page **826**. FIG. **8**B illustrates a first portion of the position details page **826** and FIG. **8**B illustrates a second portion of the position details page which has been scrolled to reveal additional details of the active stops. In various embodiments, the position details page **826** may be configured to display position specific data associated with the user account, such as, for example, all positions associated with the selected currency pair. The position details page **826** may include pair specific margin composition indicator **828**. The pair specific margin composition indicator **828** may be configured to function in a like manner as the total margin composition indicator **808**, but may make calculations and displays based only on those positions associated with the selected position details (e.g., only XBTUSD positions). The position details page **826** may include an order details frame **830** which may display details of all active orders **832**, all active stops **834**, and close on trigger conditions **842** associated with the active stops **834**.
(96) In various embodiments, the position details page **826** may be configured command the trading platform **200** to close all positions associated with the selected currency pair in response to an interaction with a close position button **836**. In response to receiving an interaction with the close position button **836**, the system may automatically generate one or more orders configured to unwind the associated position and, in response, may execute a plurality of transactions based on the orders. In like regard, the position details page **826** may be configured command the trading platform **200** to cancel all orders and/or to cancel all stops associated with the selected currency pair in response to an interaction with a corresponding cancel all orders button **838** and a cancel all stops button **840**. In response to receiving an interaction with the cancel all orders button **838**, the system may remove all active orders associated with the selected currency pair from an order book of the trading platform **200**. In response to receiving an interaction with the cancel all stops button **840**, the system may remove all active stop orders associated with the selected currency pair from an order book of the trading platform **200**. In various embodiments, the position details page **826** may be configured start an order flow process of the system. For example, the position details page **826** may include a buy button **844** and/or a sell button **846** and the system may start the order flow process in response to receiving an interaction with the buttons (**844**, **846**). In various embodiments, the order details frame **830** may be configured to receive an interaction with any of the active orders **832**, active stops **834**, and/or close on trigger conditions **842** displayed therein.
(97) In various embodiments and with additional reference to FIG. **8**D, in response to receiving an interaction with the order details frame **830** the system may display an amend position interface **848**. The amend position interface **848** may be configured to enable modification of various data associated with a discrete order displayed in the order details frame **830**. The discrete order may be determined based on the interaction, for example, the discrete order may determined based on the system registering a touch interaction with the discrete order. The amend position interface **848** may be configured to receive a size input via size input field **850** and/or a limit input via limit input field **852**. The amend position interface **848** may display current values for these inputs and, in response to receiving an interaction with either of the input fields (**850**, **852**) may display a keypad **854**. In this regard, the amend position interface **848** may facilitate altering the a current values to a new value via registering keypad **854** inputs. In various embodiments, the amend position interface **848** may prompt for a confirmation action and, in response to receiving the confirmation action,

may set the current value to the new value. For example, amend position interface **848** may display a ‘confirm amend’ slider **856** and may receive an interaction such as dragging the slider across the amend position interface **848**. In various embodiments, the amend position interface **848** may enable cancellation of the discrete order. For example, the system may receive an interaction with a cancel order button **858** and remove the associated order from the order book of the trading platform **200**.

(98) In various embodiments and with additional reference to FIGS. **9**A, **9**B, and **9**C a market data page **900** of the system **100** is illustrated. The market data page **900** may be configured to display various market data of a selected currency pair. For example, the market data page **900** may display market data related to the XBTUSD pair. In various embodiments, the market data page **900** may display data **902** such as a current price, a 24 hour percent change in price, an index price, a volume, a short gains, a funding window, an open instrument volume, a gain/loss indicator icon, and/or the like. With brief reference to FIG. **9**C, the market data page **900** is illustrated displaying order book data frame **908**. In various embodiments, the order book data frame **908** may be displayed in response to receiving an interaction with the market data page **900** such as, for example, scrolling downward or a downward swipe at the user device **104**.

(99) The market data page **900** may include a real-time graphical display of the market (i.e., market graph **904**). The market graph **904** may display various market data over time such as, for example, a spot price labeled ‘XBTUSD’ and an index price ‘.BXBT’. In various embodiments, the market graph **904** may be a candlestick chart and may be selectable across various time intervals. The market data page **900** may be configured to receive market graph time interval inputs via radio buttons **906**. As shown in FIG. **9**A, the market graph **904** is displayed at one (1) minute intervals with the ‘1 m’ radio button selected. With brief reference to FIG. **9**B, the ‘15 m’ radio button is shown selected and the corresponding alteration to the market graph **904** is illustrated. In various embodiments, the market data page **900** may be configured start an order flow process of the system **100**. For example, the market data page **900** may include a buy button **910** and/or a sell button **912** and the system may start the order flow process in response to receiving an interaction with the buttons (**910**, **912**). In various embodiments, the market data page **900** may be configured start an leverage setting process of the system **100**. For example, the market data page **900** may include a leverage setting button **914** and the system may start the leverage setting process in response to receiving an interaction with the leverage setting button **914**.

(100) In various embodiments and with additional reference to FIGS. **10**A and **10**B, the system may display a leverage setting interface **1000** in response to receiving the interaction with the leverage setting button **914**. The leverage setting interface **1000** may be configured to enable leverage settings across orders generated in the order flow process. The leverage setting interface **1000** may include a usable margin display **1002**, liquidation data, a leverage type selector **1004**, and a leverage ratio **1008**, and or the like. The leverage setting interface **1000** may be configured to receive a leverage type input via an interaction with the leverage type selector **1004** and a leverage ratio input via the keypad **1006**. In various embodiments, the usable margin display may be a bar graph. In various embodiments, the leverage setting interface **1000** may prompt for a confirmation action and, in response to receiving the confirmation action, may set the current leverage data to the new data. For example, leverage setting interface **1000** may display a ‘set leverage’ slider **1010** and may receive an interaction such as dragging the slider across the leverage setting interface **1000**.

(101) With additional reference to FIGS. **11**A through **11**F, various pages and frames of an order interface **1100** of system **100** are illustrated in accordance with various embodiments. In response to starting the order flow process, the system may display the order interface **1100**. The order interface **1100** is configured to receive various order flow inputs which define generation of a platform command. For example, the order flow interface may be configured to receive data or inputs associated with a buy order, a sell order, an order modification, a leverage setting, or a close position command. The system may generate the platform command based on these inputs. The

platform command may be received by the trading platform **200** which may, in response, execute one or more transactions based on the platform command. In various embodiments, one or more platform commands may be batched by the system or generated as a platform command set which may be executed simultaneously or sequentially by the trading platform **200**.

(102) In various embodiments, the order interface **1100** includes an order details frame **1102** which displays current order details such as, for example, a current position, a position after execution of the current order, a liquidation estimation, a calculated PnL, an estimated fill price, and/or the like. The order interface **1100** includes a limit setting button **1104** a stop setting **1106** button and an edit order button **1108**. In response to receiving an interaction with the edit order button **1108**, the system may display a prompt to edit the order quantity **1110**. For example, as shown in FIG. **11**B the system may display a keypad **1112** and receive an order quantity via the keypad **1112**. The edit order button **1108** may display 'preview' and the system may display the effect on the order details frame **1102** of the new order quantity **1110**.

(103) In response to receiving an interaction with the stop setting **1106** button, the system may display a stop settings frame **1114** shown in FIG. **11**C. The stop setting **1106** button may display an icon indicating expansion of the stop settings frame **1114**. The stop settings frame **1114** may be configured to receive order flow inputs used to define a stop order. The stop settings frame **1114** may receive and display order flow inputs such as a trigger price **1116**, a close on trigger condition switch **1118**, and a last price setting button **1120**. For example, in response to an interaction with the trigger price **1116** the system may display the keypad and receive the trigger price input. In response to receiving an interaction with the close on trigger condition switch **1118**, the system may set a close on trigger condition of the order. In various embodiments, in response to receiving an interaction with the last price setting button **1120**, the system may set the trigger price to a last filled order price of the currency pair on the exchange platform **200**.

(104) In response to receiving an interaction with the limit setting **1104** button, the system may display a limit settings frame **1122** shown in FIG. **11**D. The limit setting **1104** button may display an icon indicating expansion of the limit settings frame **1122**. The limit settings frame **1122** may be configured to receive order flow inputs used to define a limit order. The limit settings frame **1122** may receive and display order flow inputs such as a limit price **1124**, a post-only condition switch **1126**, and a reduce-only condition switch **1128**. In response to receiving an interaction with the post-only condition switch **1126**, the system may set a post-only condition of the order. In this regard, the system may generate a platform command to the trading platform **200** to accept the associated order only if it will not immediately be executed by the trading platform **200**. In response to receiving an interaction with the reduce-only condition switch **1128**, the system may set a reduce-only condition of the order.

(105) In various embodiments and as shown in FIG. **11**E, the system may display both the stop settings frame **1114** and the limit settings frame **1122** simultaneously in response to receiving interactions with both the limit setting **1104** button and stop setting **1106** button. In various embodiments, the order interface **1100** may prompt for a confirmation action and, in response to receiving the confirmation action, may generate the platform command based on the received order flow inputs. For example, order interface **1100** may display a 'sell' or 'buy' slider **1130** (which may depend on the interactions with the buy button **910** and sell button **912**) and may receive an interaction such as dragging the slider across the order interface **1100**. In response to receiving the confirmation action, the system may display a 'submitting' notification in the slider **1130** as illustrated in FIG. **11**F. In various embodiments, the system may generate one or more alerts **1132** in response to receiving the confirmation action and may prompt for a re-confirmation input. The system may receive the re-confirmation input, for example, via an interaction with a 'submit' button **1134**.

(106) In various embodiments and with additional reference to FIGS. **12**A and **12**B, a close position interface **1200** of system **100** is illustrated. The close position interface **1200** may be

configured to receive order flow inputs used to close an entire position for a selected currency pair. In this regard, the close position interface is configured to generate a platform command set controlling the trading platform to execute a plurality of transactions closing the associated positions. In various embodiments, the close position interface **1200** may display a notifications frame **1202** which describes the platform command set. The close position interface **1200** may include an add limit price button **1204**. In response to receiving an interaction with the add limit price button **1204**, the system may display a limit settings frame **1208** configured to receive a limit price input (displayed in field **1212**). The system may receive the limit price input via a keypad **1210** which may be displayed in response to receiving an interaction with the field **1212**. In various embodiments, the system may remove the limit price and revert to a market price order in response to an interaction with the remove limit price button **1214**. In various embodiments, the close position interface **1200** may prompt for a confirmation action and, in response to receiving the confirmation action, may generate the platform command set based on the received order flow inputs. For example, close position interface **1200** may display a slider **1206** and may receive an interaction such as dragging the slider across the close position interface **1200**.

(107) In various embodiments and with additional reference to FIGS. **13**A, **13**B, and **13**C an order reporting display **1300** of system **100** is illustrated. The order reporting display **1300** may comprise various framed notices which fly in from the top of a screen of the user device **104**. For example, an order submitted notification **1302** may include data or a summary of the submitted order such as, ‘close XBTUSD at Market by buying 100 contracts’. As shown in FIG. **13**B a second execution notice **1304** is flying in over a first execution notice **1306**. The execution notices (**1304**, **1306**) may include data about the fill status of the associated order, e.g., ‘68 contracts remaining’ or ‘53 contracts remaining’. As shown in FIG. **13**C, a fully filled notice **1308** may display details on a completed filled order such as price and quantity, e.g. ‘53 XBTUSD contracts bought at 10344.5’

(108) In various embodiments and with additional reference to FIGS. **14**A and **14**B, a portfolio view page **1400** of system **100** is illustrated. The portfolio view page **1400** may be configured to display the composition of a portfolio associated with the user account. For example, the portfolio view page **1400** may include one or more instrument details frames **1402** associated with the various instruments (e.g., currency pairs such as ‘XBTUSD’, contracts such as ‘XBT7D_U105’, and/or the like) comprising the portfolio. The instrument details frames **1402** may display instrument related and/or calculated data such as position size, liquidation price, entry price, ROE, PnL, value, active/open orders, stops, and/or the like. In various embodiments and in response to receiving an interaction with an instrument details frame **1402** the system may display a portfolio details view page **1404** as shown in FIG. **14**B. The portfolio details view page **1404** may include an additional details frame **1406** which may show a margin composition indicator **1408** associated with the instrument. In various embodiments, the portfolio details view page **1404** may include a close position button **1410**. The system may launch a close position process in response to receiving an interaction with the close position button **1410** and may display the close position interface **1200**.

(109) In various embodiments, and with combined reference to FIGS. **15**A through **20** an advanced position interface of platform **200** is illustrated. The advanced interface may include elements defining a long position interface **1500** and a short position interface **1600**. The advanced position interface may include a market depths chart **1502**, a buy sell toggle **1504**, a currency pair indicator **1506**, a leverage setting button **1508**, an order type display **1510**, a mark price indicator **1512**, a quantity selection button **1514**, and a quantity input frame **1516**. In various embodiments, the advanced position interface may include a dynamic margin position indicator **1700**. The dynamic margin position indicator may display a real time calculation of an existing position against a maximum position (based on a current position value and an available margin) associated with a trading account and a currency pair. The dynamic margin position indicator may display in real time the effect of a new order on the existing position and the maximum position.

(110) The dynamic margin position indicator may include a background bar **1702**. The system may calculate the background bar size based on the corresponding maximum margin value in a long direction **1712** and a short direction **1714** and display the background bar. The system may calculate a zero position **1704** based on the long direction and the short direction and display a zero indicator on the background bar. In this regard, the background bar may be scaled to indicate the maximum margin available to an account holder in the long and short direction (i.e., either side of the zero point). The system may determine a current position value and display a current position bar **1706** based on the magnitude of the current position value relative to the corresponding maximum margin value. The system may overlay the current position bar on the background bar. The system may receive order data including a currency pair and a quantity. The system may calculate a new position value based on the current position and the quantity. The system may display an arrow **1708** on the background bar indicating the new position value.
(111) In various embodiments, the arrow may be pointed toward the long or the short direction depending on the order type and the quantity. In various embodiments, the system may display a new position bar **1710** overlaid on the background bar. The new position bar may be scaled to the background bar and be displayed based on the difference between the current position and the new position. In various embodiments the new position bar may be displayed relatively brighter than the current position bar. In various embodiments, the new position bar and the current position bar may be color coded based on the long direction and the short direction. For example, if displayed in the long direction the current position bar and the new position bar may be hues and shades of green and if displayed in the short direction may be hues and shades of red. In various embodiments, the system may dynamically alter the length of the new position bar and/or the current position bar based on the order data. In this regard, the dynamic margin position indicator may indicate margin that is currently—or will be after execution of the order—assigned to the user's position. Further, the dynamic margin position indicator may thereby indicate margin that will be returned to the user's available balance after executing an order reducing a position.
(112) In various embodiments, the system may receive a request to change the quantity of an order and recalculate margin requirements based on either the last traded price, or by using the received order data and may update the dynamic margin position indicator in response. The system may receive a request to add a limit price. In response, the system may change margin calculations of the dynamic margin position indicator from an estimation based on the last traded or order book data, to an absolute price based on a limit price received in the order data. In various embodiments, the system may receive a request to change the limit price and may recalculate the margin requirements based on a new limit price. In various embodiments, the system may receive a request to add or change a trigger price. In response, the system may change margin calculations of the dynamic margin position indicator to use the trigger as an estimated execution price. The system may update the display of the dynamic margin position indicator including any of the current position bar, the new position bar, the background bar, and/or the arrow in response to recalculating the margin requirements and/or changing the margin calculations.
(113) In various embodiments, the system may receive quantity data via a quantity input interface **2000**. In various embodiments, the quantity input interface may display a keypad comprising numerical buttons, a value toggle, a decimal point button, and a backspace button. In various embodiments, the quantity input interface may include an interface type toggle **2002** configured to change the interface type between a numerical type and a percentage type. In response to a user selecting the percentage type toggle, the system may display a percentage tape scale **2004**. The tape scale may extend between a 0% position and a 100% position. The system may receive a percentage type input based on, for example, a user dragging along the tape scale to a desired position on the tape scale. The system may display a numeric indication of the selected percentage input.
(114) With reference to FIG. **21**, a launch process **2100** of a system **100** is illustrated in accordance

with various embodiments. The system may receive a launch command including a sign-in data. The system may determine an operational mode based on the sign in data **2102**. Where the sign-in data does not comprise authentication data, the system may start an anonymous mode process **2104**. In response, the system may start an authentication setup process **2106** which may generate an authentication request based on the sign-in data. In various embodiments, in response to receiving an authentication data **2108** the system may start a trading interface process **2110**. The system may display a trading interface comprising a portfolio drawer. The portfolio drawer display may display portfolio information associated with the authentication data and/or sign in data.

(115) In various embodiments and with additional reference to FIG. **22**, a market view page process **2200** of system **100** is illustrated. In response to receiving the authentication data **2108** the system may display a splash screen **2202** and may then display a market view page **2204**. The system may initiate a connection **2206** with trading platform **200** and receive data from the trading platform **200**. For example, the system may populate the market view page **2204** with a cryptocoin list **2208** and/or a disabled state cryptocoin list **2210**. In various embodiments, the market view page **2204** may be configured to receive one or more inputs from the user device **104**. For example, the market view page **2204** may be configured to receive an account and settings section **2212**, a go to instrument selection **2214**, and a go to portfolio selection **2216**.

(116) In various embodiments and with additional reference to FIG. **23** a leverage setting page process **2300** of system **100** is illustrated. In response to receiving an initiate set leverage input **2302** from a user device **104**. The system may display a set leverage page **2304** such as, for example, leverage setting interface **1000**. The set leverage page **2304** may be configured to receive and/or display one or more inputs to initialize leverage associated input states such as, for example, a leverage isolated input **2306**, a leverage cross input **2308**, and a leverage receipt preview **2310**. In response to initializing the leverage associated input states, the system may display one of an instrument details **2312** page or an order modification page **2314**.

(117) In various embodiments and with additional reference to FIG. **24** a second launch process **2400** of system **100** is illustrated. User device **104** may receive a command **2402** to launch the native application **110**. In response the system may determine whether the native application **110** is cached locally on the user device **104** (**2404**). The system may display a splash screen **2406** in response to determining an uncashed condition and proceed to cache the native application **110**. Otherwise the system may determine a sign-in state (**2408**). In response to a failed sign-in state, the system may proceed to an anonymous mode process **2410**. Otherwise, the system may proceed to an authentication process **2412** and display an authentication page (e.g., authentication interface **510**). In various embodiments, the system may determine whether authentication is restricted **2414** and drop to a restricted process **2416** which may force a logout event **2418** and return to the anonymous mode process **2410**. If authentication is not restricted, the system may perform a deep link check **2420**. In response, the system may launch a market and portfolio view process **2422** or a create/amend order process **2424**.

(118) In various embodiments and with additional reference to FIG. **25**, an instrument details page process **2500** of system **100** is illustrated. The system may receive a view instrument action **2502** and, in response, display an instrument details page **2504**. The instrument details page **2504** may be configured to generate an initiate set leverage action **2506**, initiate buy/sell order action **2508**, or go to portfolio action **2510**. The system may receive an interaction with a price chart **2512** or an interaction with an orderbook **2514** via the instrument details page **2504**. In response the system may start an order flow process. For example, the system may generate an initiate draft buy/sell order with size action **2516** or may generate an initiate draft buy/sell order with price action **2518**.

(119) In various embodiments and with additional reference to FIG. **26**, an close position page process **2600** of system **100** is illustrated. The system may receive an initiate close position action **2602** and in response display a close position page **2604** (e.g., close position interface **1200**). The process may initialize to a close at market state **2606** and may determine a transition to close at

limit price **2608** based on receiving a state selection. The system may receive a limit price input **2610**. The system may determine whether the limit price input is entered (i.e., non-zero value) **2612**. In response to determine an entered limit price, the system may await confirmation of a limit close state **2614**. Otherwise the system will await confirmation of a market close state **2616**.

(120) In various embodiments and with additional reference to FIG. **27** a third launch process **2700** of system **100** is illustrated. Process **2700** includes steps, systems, and features of process **2400**, but incorporates a restricted mode process. In response to determining a sign-in state **2702**, the system may perform a restricted authentication soft check **2704**. In response to the a restricted authentication soft check, the system may set a restricted warning state **2710** or a default state **2712** of authentication process **2708**. In various embodiments, the system may perform a hard check of whether authentication is restricted **2714**. The system may drop to a restricted process **2716** in response to a restricted result from the hard check. In various embodiments, the restricted process **2716** may enable a restricted mode **2718** of the market and portfolio view process.

(121) In various embodiments and with additional reference to FIG. **28**, a portfolio drawer process **2800** of system **100** is illustrated. The system may display a portfolio drawer page **2802** in response to an interaction with a market view page **2804**, an accounts and settings page **2806**, and/or an instrument details page **2808**. The system may display an overview page **2810** of the portfolio drawer page **2802** in response to the interactions with the market view page **2804** and the accounts and settings page **2806**. The system may display an instrument's portfolio details page **2812** in response to the interaction with the instrument details page **2808**. The overview page **2810** may be configured to generate a view instrument details action **2814**. In response to generating the view instrument details action **2814** the system may display the instrument details page **2808**. In various embodiments, the instrument's portfolio details page **2812** may be configured to generate an initiate close position action **2816**, an amend order or stop action **2818**, and the view instrument details action **2814**. In various embodiments, the system may display the close position interface in response to the initiate close position action **2816** and may display the order interface in response to the amend order or stop action **2818**.

(122) In various embodiments and with additional reference to FIGS. **29**A and **29**B, an order modification process **2900** of system **100** is illustrated. In various embodiments, the order modification process may be initiated in response to receiving any of an initiate draft buy/sell order action **2902**, a initiate draft buy/sell order with size action **2904**, or an initiate draft buy/sell order with price action **2906**. The system may display an order modification page **2908** in response. The system may receive a size input **2910**. The system may check for a limit price setting **2912** and may check for a stop trigger setting **2914**. In response to the limit price setting **2912**, the system may await a limit price input **2916**. In response to the stop trigger setting **2914**, the system may await a stop trigger input **2918** and a trigger type input **2920**. In response to receiving the inputs (**2916**, **2918**, **2920**) the system may check for an entered limit price **2922** and, if none, may check for a trigger price entered **2924**. If no trigger price entered, the system may display a market order receipt preview **2926** and generate a confirm buy/sell order action **2928**.

(123) If a trigger price is entered, the system may execute a direction finding process **2930** based on the trigger price input and the trigger type input. The system may display a take profit market order receipt preview **2932** or a stop market order receipt preview **2934** based on the outcome of direction finding process **2930** and proceed to generate the confirm buy/sell order action **2928**. If a limit price is entered the system may determine a post only option **2936** and, if no trigger price is entered, the system may display a limit order receipt preview **2938** and proceed to generate the confirm buy/sell order action **2928**. If a limit price and a trigger price are entered, the system may execute a direction finding process **2930** based on the trigger price input and the trigger type input. The system may display a take profit limit order receipt preview **2940** or a stop limit order receipt preview **2942** based on the outcome of direction finding process **2930** and proceed to generate the confirm buy/sell order action **2928**.

(124) In various embodiments and with additional reference to FIGS. **30**A and **30**B, a maintenance process **3000** of system **100** is illustrated. Native application **110** may be launched on user device **104** and determine a connection to a wide area network **3002**. In response to determining a connection failure, the system may enter a not connected state **3004** and display a disconnected status. In response to entering the not connected state **3004**, the system may start a reconnection process **3006**. In response to determining a connected state, the system may check for updates **3008**. In response to determining an available update the system may enter a forced update state **3010** and await an update now action **3012**. Otherwise, the system may check for a server maintenance state **3014**. In response to determining active server maintenance, the system may enter a server maintenance state **3016** and generate a status page punch-out action **3018**. In response to the status page punch-out action **3018**, the system may launch **3020** a web browser of the user device **104**. While in the server maintenance state **3016** the system may check for completed server maintenance **3022**. In response to determining completed server maintenance the system may check for sign-in process state **3024**. If sign in is not in process the system may check for a signed-in state **3026**. If the signed-in state is true then the system proceeds to check for prior state data of the native app **110** (**3028**). If prior state data exists, the system may determine whether the prior session is still live **3030**. If the prior session is still live then the process progresses to open the prior state **3032**.
(125) In various embodiments and with additional reference to FIGS. **31**A and **31**B, a sign-in process **3100** of system **100** is illustrated. The system may receive a sign-in action **3102** and display a sign in screen **3104** in response. The system may receive an email and a password **3106**. The system may receive an authentication token **3108**. The system may determine a success state of the sign-in based on the email and the password **3110**. In response to a failed success state, the system may display an error message **3112** and return to the sign in screen **3104**. Otherwise, the system may progress to determine receipt of the authentication token **3114**. If no token is entered, the system may progress to a five (5) minute wait state **3116** for entry of the authentication token. The system may exit the wait state **3115** to the sign in screen **3104**. In response to receiving the authentication token the system may determine a mobile device verification state **3118**. In response to determining an unverified device, the system may launch a device verification subprocess **3120**. Otherwise the system may progress to a set new passcode prompt **3122**. The system may receive the passcode and prompt for reentry of the passcode **3124**. The system may compare the passcodes to determine a match **3126**. The system may return an error in response to a mismatch **3128**. Otherwise, the system may determine available biometric authentication **3130**. If biometric authentication is unavailable, the system may progress to display the home screen **3132**. Otherwise, the system may check if touch ID is available **3134**. If true, the system may start a biometric authentication process **3136**. Else, the system may prompt for permissions to activate touch ID **3138** and either start the biometric authentication process or disable the biometric authentication process based on the permissions.
(126) In various embodiments, in response to launching the device verification subprocess **3120** the system may transmit a verification link for mobile device verification **3140**. The system may enter a 30 minute wait state **3142** while awaiting an interaction with the verification link. While in the wait state **3142**, the system may retransmit the verification link **3144**. The system may determine whether the retransmission was successful **3146** and, if not, may display an error message **3148**. The system may determine an interaction with the verification link **3150**. The system may determine whether the interaction with the verification link is via the mobile device **104** (**3152**). If false, the system may generate a web-browser punch out action **3158**. In response to the punch out action, the system may display an error message **3160**. If true, the system may launch the native app **110** of the mobile device **104** (**3154**). The system may check if the verification link is expired **3156**. If true, the system may display a verification link expired message **3162** and recycle to transmit a new verification link. Else, the system may determine the device verified **3164**. If false,

the system may generate a verification error message **3166**.

(127) In various embodiments and with additional reference to FIGS. **32**A and **32**B, a biometric authentication process **3200** of system **100** is illustrated. The system may determine a biometric authentication enabled state **3202**. If false, the system may determine a passcode enabled state **3204**. If false, the system may enter a recycle process **3206** and recycle to a previous state or progress to the home screen. In response to determining a true biometric authentication enabled state, the system may check for a device specific authentication state such as a face ID state **3208**, a touch ID state **3210**, or a fingerprint ID state **3212**. In response, the system may launch a corresponding device specific biometric authentication process. For example, face ID authentication process **3214**, touch ID authentication process **3216**, and fingerprint authentication process **3218**. Any of the authentication processes **3214**, **3216**, **3218** may return a biometric validated state **3220** and the system may progress to the recycle process **3206**. In various embodiments, in response to a passcode enabled state **3222** any of the processes **3214**, **3216**, **3218** may prompt for entry of a passcode **3224**. They system may receive the passcode and may determine a validated passcode state **3226**. In response to the validated passcode state and the biometric validated state the system may progress to the recycle process **3206**. Else, the system may progress to a failed attempt process **3228** and increment a failed attempt state counter **3230**. In response to the failed attempt state counter exceed a threshold, the system may force a sign-out event **3232**. In response, the system may display an error message **3234** and return to the sign-in screen. In various embodiments, any of the processes **3214**, **3216**, **3218** may return a biometric failed state **3236**. In response the system may prompt for entry of the passcode **3224**.

(128) Via use of the interface, processes, and techniques disclosed in FIGS. **1** through **32**, a user of the system may more readily visualize and understand the effect of proposed transactions. In this manner, the user can make quicker and/or better informed decisions regarding the proposed transactions, resulting in transactions that more closely track the user's desired outcomes (for example, by reducing and/or eliminating delays in entering transaction details, reducing and/or eliminating delays associated with user consideration or evaluation of a proposed transaction, and/or the like). Moreover, by reducing the amount of time a mobile application embodying and/or utilizing the system may need to be operative on a mobile device in order to effectuate a desired transaction, device battery life may be improved, network congestion may be reduced, and/or other improvements to operation of the mobile device or other computing or communications systems may be realized.

(129) The detailed description of various embodiments herein makes reference to the accompanying drawings and pictures, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that logical and communicative changes may be made without departing from the spirit and scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. For example, the steps recited in any of the method or process descriptions may be executed in any suitable order and are not limited to the order presented. Moreover, certain of the functions or steps may be outsourced to or performed by one or more third parties. Modifications, additions, or omissions may be made to the systems, apparatuses, and methods described herein without departing from the scope of the disclosure. For example, the components of the systems and apparatuses may be integrated or separated. Moreover, the operations of the systems and apparatuses disclosed herein may be performed by more, fewer, or other components and the methods described may include more, fewer, or other steps. Additionally, steps may be performed in any suitable order. As used in this document, “each” refers to each member of a set or each member of a subset of a set. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component may include a singular embodiment. Although specific advantages have been enumerated herein, various embodiments may include

some, none, or all of the enumerated advantages.

(130) Systems, methods, and computer program products are provided. In the detailed description herein, references to "various embodiments," "one embodiment," "an embodiment," "an example embodiment," etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(131) As used herein, "satisfy," "meet," "match," "associated with", or similar phrases may include an identical match, a partial match, meeting certain criteria, matching a subset of data, a correlation, satisfying certain criteria, a correspondence, an association, an algorithmic relationship, and/or the like. Similarly, as used herein, "authenticate" or similar terms may include an exact authentication, a partial authentication, authenticating a subset of data, a correspondence, satisfying certain criteria, an association, an algorithmic relationship, and/or the like.

(132) Terms and phrases similar to "associate" and/or "associating" may include tagging, flagging, correlating, using a look-up table or any other method or system for indicating or creating a relationship between elements, such as, for example: (i) a trading account and (ii) an order (e.g., contract, security, future) and/or digital channel. Moreover, the associating may occur at any point, in response to any suitable action, event, or period of time. The associating may occur at predetermined intervals, periodic, randomly, once, more than once, or in response to a suitable request or action. Any of the information may be distributed and/or accessed via any suitable method, for example a software enabled link, wherein the link may be sent via an email, text, post, social network input, and/or any other method known in the art.

(133) The process flows and screenshots depicted in the figures are merely embodiments and are not intended to limit the scope of the disclosure. For example, the steps recited in any of the method or process descriptions may be executed in any suitable order and are not limited to the order presented. It will be appreciated that the following description makes appropriate references not only to the steps and user interface elements depicted in the figures, but also to the various system components as described above with reference to FIG. **1**. It should be understood at the outset that, although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described below. Unless otherwise specifically noted, articles depicted in the drawings are not necessarily drawn to scale.

(134) Computer programs (also referred to as computer control logic) may be stored in main memory and/or secondary memory. Computer programs may also be received via communications interface. Such computer programs, when executed, enable the computer system to perform the features as discussed herein. In particular, the computer programs, when executed, enable the processor to perform the features of various embodiments. Accordingly, such computer programs represent controllers of the computer system.

(135) These computer program instructions may be loaded onto a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions that execute on the computer or other programmable data processing apparatus create means for implementing the functions specified in the flowchart block or blocks. These computer program instructions may also be stored in a computer-readable memory that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable memory produce an article of

manufacture including instruction means which implement the function specified in the flowchart block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions which execute on the computer or other programmable apparatus provide steps for implementing the functions specified in the flowchart block or blocks.

(136) In various embodiments, software may be stored in a computer program product and loaded into a computer system using removable storage drive, hard disk drive, or communications interface. The control logic (software), when executed by the processor, causes the processor to perform the functions of various embodiments as described herein. In various embodiments, hardware components may take the form of application specific integrated circuits (ASICs). Implementation of the hardware state machine so as to perform the functions described herein will be apparent to persons skilled in the relevant art(s).

(137) As will be appreciated by one of ordinary skill in the art, the system may be embodied as a customization of an existing system, an add-on product, a processing apparatus executing upgraded software, a stand-alone system, a distributed system, a method, a data processing system, a device for data processing, and/or a computer program product. Accordingly, any portion of the system or a module may take the form of a processing apparatus executing code, an internet-based embodiment, an entirely hardware embodiment, or an embodiment combining aspects of the internet, software, and hardware. Furthermore, the system may take the form of a computer program product on a computer-readable storage medium having computer-readable program code means embodied in the storage medium. Any suitable computer-readable storage medium may be utilized, including hard disks, solid-state storage devices, optical storage devices, magnetic storage devices, and/or the like.

(138) In various embodiments, components, modules, and/or engines of system **100** may be implemented as micro-applications or micro-apps. Micro-apps are typically deployed in the context of a mobile operating system, including for example, Windows mobile, Android, Apple iOS, Blackberry, and the like. The micro-app may be configured to leverage the resources of the larger operating system and associated hardware via a set of predetermined rules which govern the operations of various operating systems and hardware resources. For example, where a micro-app desires to communicate with a device or network other than the mobile device or mobile operating system, the micro-app may leverage the communication protocol of the operating system and associated device hardware under the predetermined rules of the mobile operating system. Moreover, where the micro-app desires an input from a user, the micro-app may be configured to request a response from the operating system which monitors various hardware components and then communicates a detected input from the hardware to the micro-app.

(139) The system and method may be described herein in terms of functional block components, screen shots, optional selections, and various processing steps. It should be appreciated that such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions. For example, the system may employ various integrated circuit components, e.g., memory elements, processing elements, logic elements, look-up tables, and the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly, the software elements of the system may be implemented with any programming or scripting language, for example such as C, C++, C#, Java, Javascript, Javascript Object Notation (JSON), VBScript, Macromedia Cold Fusion, Cobol, active server pages, Perl, assembly, PHP, awk, Python, Visual Basic, SQL Stored Procedures, PL/SQL, any Unix shell script, and/or extensible markup language (XML) with the various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements. Further, it should be noted that the system may employ any number of conventional techniques for data transmission, signaling, data processing, network control, and the

like. Still further, the system could be used to detect or prevent security issues with a client-side scripting language, such as Javascript, VBScript, or the like.

(140) The system and method are described herein with reference to screen shots, block diagrams and flowchart illustrations of methods, apparatus, and computer program products according to various embodiments. It will be understood that each functional block of the block diagrams and the flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, respectively, can be implemented by computer program instructions.

(141) Accordingly, functional blocks of the block diagrams and flowchart illustrations support combinations of means for performing the specified functions, combinations of steps for performing the specified functions, and program instruction means for performing the specified functions. It will also be understood that each functional block of the block diagrams and flowchart illustrations, and combinations of functional blocks in the block diagrams and flowchart illustrations, can be implemented by either special purpose hardware-based computer systems which perform the specified functions or steps, or suitable combinations of special purpose hardware and computer instructions. Further, illustrations of the process flows and the descriptions thereof may make reference to user windows applications, webpages, websites, web forms, prompts, etc. Practitioners will appreciate that the illustrated steps described herein may comprise, in any number of configurations, including the use of windows applications, webpages, web forms, popup windows applications, prompts, and the like. It should be further appreciated that the multiple steps as illustrated and described may be combined into single webpages and/or windows applications but have been expanded for the sake of simplicity. In other cases, steps illustrated and described as single process steps may be separated into multiple webpages and/or applications but have been combined for simplicity.

(142) In various embodiments, the software elements of the system may also be implemented using a Javascript run-time environment configured to execute Javascript code outside of a web browser. For example, the software elements of the system may also be implemented using Node.js components. Node.js programs may implement several modules to handle various core functionalities. For example, a package management module, such as NPM, may be implemented as an open source library to aid in organizing the installation and management of third-party Node.js programs. Node.js programs may also implement a process manager, such as, for example, Parallel Multithreaded Machine (“PM2”); a resource and performance monitoring tool, such as, for example, Node Application Metrics (“appmetrics”); a library module for building user interfaces, and/or any other suitable and/or desired module.

(143) Middleware may include any hardware and/or software suitably configured to facilitate communications and/or process transactions between disparate computing systems. Middleware components are commercially available and known in the art. Middleware may be implemented through commercially available hardware and/or software, through custom hardware and/or software components, or through a combination thereof. Middleware may reside in a variety of configurations and may exist as a standalone system or may be a software component residing on the internet server. Middleware may be configured to process transactions between the various components of an application server and any number of internal or external systems for any of the purposes disclosed herein. WebSphere MQTM (formerly MQSeries) by IBM, Inc. (Armonk, N.Y.) is an example of a commercially available middleware product. An Enterprise Service Bus (“ESB”) application is another example of middleware.

(144) The computers discussed herein may provide a suitable website or other internet-based graphical user interface which is accessible by users. In one embodiment, Microsoft company's Internet Information Services (IIS), Transaction Server (MTS) service, and an SQL Server database, are used in conjunction with Microsoft operating systems, Windows web server software, and Microsoft Commerce Server. Additionally, components such as Access software, SQL Server database, Oracle software, Sybase software, Informix software, MySQL software, Interbase

software, etc., may be used to provide an Active Data Object (ADO) compliant database management system. In one embodiment, the Apache web server is used in conjunction with a Linux operating system, a MySQL database, and Perl, PHP, Ruby, and/or Python programming languages.

(145) For the sake of brevity, conventional data networking, application development, and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system.

(146) In various embodiments, the methods described herein are implemented using the various particular machines described herein. The methods described herein may be implemented using the below particular machines, and those hereinafter developed, in any suitable combination, as would be appreciated immediately by one skilled in the art. Further, as is unambiguous from this disclosure, the methods described herein may result in various transformations of certain articles.

(147) The various system components discussed herein may include one or more of the following: a host server or other computing systems including a processor for processing digital data; a memory coupled to the processor for storing digital data; an input digitizer coupled to the processor for inputting digital data; an application program stored in the memory and accessible by the processor for directing processing of digital data by the processor; a display device coupled to the processor and memory for displaying information derived from digital data processed by the processor; and a plurality of databases. Various databases used herein may include: client data; merchant data; financial institution data; and/or like data useful in the operation of the system. As those skilled in the art will appreciate, user computer may include an operating system (e.g., Windows, Linux, Unix, Solaris, MacOS, etc.) as well as various conventional support software and drivers typically associated with computers.

(148) The present system or any part(s) or function(s) thereof may be implemented using hardware, software, or a combination thereof and may be implemented in one or more computer systems or other processing systems. However, the manipulations performed by embodiments may be referred to in terms, such as matching or selecting, which are commonly associated with mental operations performed by a human operator. No such capability of a human operator is necessary, or desirable, in most cases, in any of the operations described herein. Rather, the operations may be machine operations or any of the operations may be conducted or enhanced by artificial intelligence (AI) or machine learning. AI may refer generally to the study of agents (e.g., machines, computer-based systems, etc.) that perceive the world around them, form plans, and make decisions to achieve their goals. Foundations of AI include mathematics, logic, philosophy, probability, linguistics, neuroscience, and decision theory. Many fields fall under the umbrella of AI, such as computer vision, robotics, machine learning, and natural language processing. Useful machines for performing the various embodiments include general purpose digital computers or similar devices.

(149) In various embodiments, the embodiments are directed toward one or more computer systems capable of carrying out the functionalities described herein. The computer system includes one or more processors. The processor is connected to a communication infrastructure (e.g., a communications bus, cross-over bar, network, etc.). Various software embodiments are described in terms of this exemplary computer system. After reading this description, it will become apparent to a person skilled in the relevant art(s) how to implement various embodiments using other computer systems and/or architectures. The computer system can include a display interface that forwards graphics, text, and other data from the communication infrastructure (or from a frame buffer not shown) for display on a display unit.

(150) The computer system also includes a main memory, such as random access memory (RAM), and may also include a secondary memory. The secondary memory may include, for example, a

hard disk drive, a solid-state drive, and/or a removable storage drive. The removable storage drive reads from and/or writes to a removable storage unit in a well-known manner. As will be appreciated, the removable storage unit includes a computer usable storage medium having stored therein computer software and/or data.

(151) In various embodiments, secondary memory may include other similar devices for allowing computer programs or other instructions to be loaded into a computer system. Such devices may include, for example, a removable storage unit and an interface. Examples of such may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an erasable programmable read only memory (EPROM), programmable read only memory (PROM)) and associated socket, or other removable storage units and interfaces, which allow software and data to be transferred from the removable storage unit to a computer system.

(152) The terms “computer program medium,” “computer usable medium,” and “computer readable medium” are used to generally refer to media such as removable storage drive and a hard disk installed in hard disk drive. These computer program products provide software to a computer system.

(153) The computer system may also include a communications interface. A communications interface allows software and data to be transferred between the computer system and external devices. Examples of communications interface may include a modem, a network interface (such as an Ethernet card), a communications port, etc. Software and data transferred via the communications interface are in the form of signals which may be electronic, electromagnetic, optical, or other signals capable of being received by communications interface. These signals are provided to communications interface via a communications path (e.g., channel). This channel carries signals and may be implemented using wire, cable, fiber optics, a telephone line, a cellular link, a radio frequency (RF) link, wireless and other communications channels.

(154) As used herein an “identifier” may be any suitable identifier that uniquely identifies an item. For example, the identifier may be a globally unique identifier (“GUID”). The GUID may be an identifier created and/or implemented under the universally unique identifier standard. Moreover, the GUID may be stored as 128-bit value that can be displayed as 32 hexadecimal digits. The identifier may also include a major number, and a minor number. The major number and minor number may each be 16-bit integers.

(155) In various embodiments, the server may include application servers (e.g., Websphere, Weblogic, jBoss, Postgres Plus Advanced Server, etc.). In various embodiments, the server may include web servers (e.g., Apache, IIS, Google Web Server, and/or the like).

(156) A web client includes any device or software which communicates via any network, such as, for example any device or software discussed herein. The web client may include internet browsing software installed within a computing unit or system to conduct online transactions and/or communications. These computing units or systems may take the form of a computer or set of computers, although other types of computing units or systems may be used, including personal computers, laptops, notebooks, tablets, smart phones, cellular phones, personal digital assistants, servers, pooled servers, mainframe computers, distributed computing clusters, kiosks, terminals, point of sale (POS) devices or terminals, televisions, or any other device capable of receiving data over a network. The web client may include an operating system as well as various conventional support software and drivers typically associated with computers. The web-client may also run Microsoft Edge, Internet Explorer, Mozilla Firefox, Google Chrome, Apple Safari, or any other of the myriad software packages available for browsing the internet.

(157) As those skilled in the art will appreciate, the web client may or may not be in direct contact with the server (e.g., application server, web server, etc., as discussed herein). For example, the web client may access the services of the server through another server and/or hardware component, which may have a direct or indirect connection to an internet server. For example, the

web client may communicate with the server via a load balancer. In various embodiments, web client access is through a network or the internet through a commercially-available web-browser software package. In that regard, the web client may be in a home or business environment with access to the network or the internet. The web client may implement security protocols such as Secure Sockets Layer (SSL) and Transport Layer Security (TLS). A web client may implement several application layer protocols including HTTP, HTTPS, FTP, and SFTP.

(158) The various system components may be independently, separately, or collectively suitably coupled to the network via data links which includes, for example, a connection to an Internet Service Provider (ISP) over the local loop as is typically used in connection with standard modem communication, cable modem, ISDN, Digital Subscriber Line (DSL), or various wireless communication methods. It is noted that the network may be implemented as other types of networks, such as an interactive television (ITV) network. Moreover, the system contemplates the use, sale, or distribution of any goods, services, or information over any network having similar functionality described herein.

(159) The system contemplates uses in association with web services, utility computing, pervasive and individualized computing, security and identity solutions, autonomic computing, cloud computing, commodity computing, mobility and wireless solutions, open source, biometrics, grid computing, and/or mesh computing.

(160) Any of the communications, inputs, storage, databases or displays discussed herein may be facilitated through a website having web pages. The term “web page” as it is used herein is not meant to limit the type of documents and applications that might be used to interact with the user. For example, a typical website might include, in addition to standard HTML documents, various forms, Java applets, Javascript programs, active server pages (ASP), common gateway interface scripts (CGI), extensible markup language (XML), dynamic HTML, cascading style sheets (CSS), AJAX (Asynchronous JAVASCRIPT And XML) programs, helper applications, plug-ins, and the like. A server may include a web service that receives a request from a web server, the request including a URL and an IP address (e.g., 192.168.1.1). The web server retrieves the appropriate web pages and sends the data or applications for the web pages to the IP address. Web services are applications that are capable of interacting with other applications over a communications means, such as the internet. Web services are typically based on standards or protocols such as XML, SOAP, AJAX, WSDL and UDDI. Web services methods are well known in the art, and are covered in many standard texts. For example, representational state transfer (REST), or RESTful, web services may provide one way of enabling interoperability between applications.

(161) The computing unit of the web client may be further equipped with an internet browser connected to the internet or an intranet using standard dial-up, cable, DSL, or any other internet protocol known in the art. Transactions originating at a web client may pass through a firewall in order to prevent unauthorized access from users of other networks. Further, additional firewalls may be deployed between the varying components of CMS to further enhance security

(162) Encryption may be performed by way of any of the techniques now available in the art or which may become available—e.g., Twofish, RSA, El Gamal, Schorr signature, DSA, PGP, PKI, GPG (GnuPG), HPE Format-Preserving Encryption (FPE), Voltage, Triple DES, Blowfish, AES, MD5, HMAC, IDEA, RC6, and symmetric and asymmetric cryptosystems. The systems and methods may also incorporate SHA series cryptographic methods, elliptic curve cryptography (e.g., ECC, ECDH, ECDSA, etc.), and/or other post-quantum cryptography algorithms under development.

(163) The firewall may include any hardware and/or software suitably configured to protect CMS components and/or enterprise computing resources from users of other networks. Further, a firewall may be configured to limit or restrict access to various systems and components behind the firewall for web clients connecting through a web server. Firewall may reside in varying configurations including Stateful Inspection, Proxy based, access control lists, and Packet Filtering among others.

Firewall may be integrated within a web server or any other CMS components or may further reside as a separate entity. A firewall may implement network address translation ("NAT") and/or network address port translation ("NAPT"). A firewall may accommodate various tunneling protocols to facilitate secure communications, such as those used in virtual private networking. A firewall may implement a demilitarized zone ("DMZ") to facilitate communications with a public network such as the internet. A firewall may be integrated as software within an internet server or any other application server components, reside within another computing device, or take the form of a standalone hardware component.

(164) Any databases discussed herein may include relational, hierarchical, graphical, blockchain, object-oriented structure, and/or any other database configurations. Any database may also include a flat file structure wherein data may be stored in a single file in the form of rows and columns, with no structure for indexing and no structural relationships between records. For example, a flat file structure may include a delimited text file, a CSV (comma-separated values) file, and/or any other suitable flat file structure. Common database products that may be used to implement the databases include DB2 by IBM (Armonk, N.Y.), various database products available from Oracle Corporation (Redwood Shores, Calif.), Microsoft Access or SQ1 Server by Microsoft Corporation (Redmond, Wash.), MySQL by MySQL AB (Uppsala, Sweden), MongoDB, Redis, Apache Cassandra, hBase by Apache, MapR-DB by the MAPR corporation, or any other suitable database product. Moreover, any database may be organized in any suitable manner, for example, as data tables or lookup tables. Each record may be a single file, a series of files, a linked series of data fields, or any other data structure.

(165) As used herein, big data may refer to partially or fully structured, semi-structured, or unstructured data sets including millions of rows and hundreds of thousands of columns. A big data set may be compiled, for example, from a history of transactions over time, from web registrations, from social media, from records of charge (ROC), from summaries of charges (SOC), from internal data, or from other suitable sources. Big data sets may be compiled without descriptive metadata such as column types, counts, percentiles, or other interpretive-aid data points.

(166) Association of certain data may be accomplished through any desired data association technique such as those known or practiced in the art. For example, the association may be accomplished either manually or automatically. Automatic association techniques may include, for example, a database search, a database merge, GREP, AGREP, SQL, using a key field in the tables to speed searches, sequential searches through all the tables and files, sorting records in the file according to a known order to simplify lookup, and/or the like. The association step may be accomplished by a database merge function, for example, using a "key field" in pre-selected databases or data sectors. Various database tuning steps are contemplated to optimize database performance. For example, frequently used files such as indexes may be placed on separate file systems to reduce In/Out ("I/O") bottlenecks.

(167) More particularly, a "key field" partitions the database according to the high-level class of objects defined by the key field. For example, certain types of data may be designated as a key field in a plurality of related data tables and the data tables may then be linked on the basis of the type of data in the key field. The data corresponding to the key field in each of the linked data tables is preferably the same or of the same type. However, data tables having similar, though not identical, data in the key fields may also be linked by using AGREP, for example. In accordance with one embodiment, any suitable data storage technique may be utilized to store data without a standard format. Data sets may be stored using any suitable technique, including, for example, storing individual files using an ISO/IEC 7816-4 file structure; implementing a domain whereby a dedicated file is selected that exposes one or more elementary files containing one or more data sets; using data sets stored in individual files using a hierarchical filing system; data sets stored as records in a single file (including compression, SQL accessible, hashed via one or more keys, numeric, alphabetical by first tuple, etc.); data stored as Binary Large Object (BLOB); data stored

as ungrouped data elements encoded using ISO/IEC 7816-6 data elements; data stored as ungrouped data elements encoded using ISO/IEC Abstract Syntax Notation (ASN.1) as in ISO/IEC 8824 and 8825; other proprietary techniques that may include fractal compression methods, image compression methods, etc.

(168) In various embodiments, the ability to store a wide variety of information in different formats is facilitated by storing the information as a BLOB. Thus, any binary information can be stored in a storage space associated with a data set. As discussed above, the binary information may be stored in association with the system or external to but affiliated with system. The BLOB method may store data sets as ungrouped data elements formatted as a block of binary via a fixed memory offset using either fixed storage allocation, circular queue techniques, or best practices with respect to memory management (e.g., paged memory, least recently used, etc.). By using BLOB methods, the ability to store various data sets that have different formats facilitates the storage of data, in the database or associated with the system, by multiple and unrelated owners of the data sets. For example, a first data set which may be stored may be provided by a first party, a second data set which may be stored may be provided by an unrelated second party, and yet a third data set which may be stored, may be provided by an third party unrelated to the first and second party. Each of these three exemplary data sets may contain different information that is stored using different data storage formats and/or techniques. Further, each data set may contain subsets of data that also may be distinct from other sub sets.

(169) The data set annotation may also be used for other types of status information as well as various other purposes. For example, the data set annotation may include security information establishing access levels. The access levels may, for example, be configured to permit only certain individuals, levels of employees, companies, or other entities to access data sets, or to permit access to specific data sets based on the transaction, instrument, contract details, issuer, buyer, seller, user, or the like. Furthermore, the security information may restrict/permit only certain actions such as accessing, modifying, and/or deleting data sets. In one example, the data set annotation indicates that only the data set owner or the user are permitted to delete a data set, various identified users may be permitted to access the data set for reading, and others are altogether excluded from accessing the data set. However, other access restriction parameters may also be used allowing various entities to access a data set with various permission levels as appropriate.

(170) One skilled in the art will also appreciate that, for security reasons, any databases, systems, devices, servers, or other components of the system may consist of any combination thereof at a single location or at multiple locations, wherein each database or system includes any of various suitable security features, such as firewalls, access codes, encryption, decryption, compression, decompression, and/or the like.

(171) Practitioners will also appreciate that there are a number of methods for displaying data within a browser-based document. Data may be represented as standard text or within a fixed list, scrollable list, drop-down list, editable text field, fixed text field, pop-up window, and the like. Likewise, there are a number of methods available for modifying data in a web page such as, for example, free text entry using a keyboard, selection of menu items, check boxes, option boxes, and the like.

(172) As used herein, the term “network” includes any cloud, cloud computing system, or electronic communications system or method which incorporates hardware and/or software components. Communication among the parties may be accomplished through any suitable communication channels, such as, for example, a telephone network, an extranet, an intranet, internet, point of interaction device (point of sale device, personal digital assistant, smartphone, kiosk, etc.), online communications, satellite communications, off-line communications, wireless communications, transponder communications, local area network (LAN), wide area network (WAN), virtual private network (VPN), networked or linked devices, keyboard, mouse, and/or any

suitable communication or data input modality. Moreover, although the system is frequently described herein as being implemented with TCP/IP communications protocols, the system may also be implemented using IPX, AppleTalk, IP-6, NetBIOS, OSI, any tunneling protocol (e.g. IPsec, SSH, etc.), or any number of existing or future protocols. If the network is in the nature of a public network, such as the internet, it may be advantageous to presume the network to be insecure and open to eavesdroppers. Specific information related to the protocols, standards, and application software utilized in connection with the internet is generally known to those skilled in the art and, as such, need not be detailed herein.

(173) “Cloud” or “Cloud computing” includes a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud computing may include location-independent computing, whereby shared servers provide resources, software, and data to computers and other devices on demand.

(174) As used herein, “transmit” may include sending electronic data from one system component to another over a network connection. Additionally, as used herein, “data” may include encompassing information such as commands, queries, files, data for storage, and the like in digital or any other form.

(175) The term “non-transitory” is to be understood to remove only propagating transitory signals per se from the claim scope and does not relinquish rights to all standard computer-readable media that are not only propagating transitory signals per se. Stated another way, the meaning of the term “non-transitory computer-readable medium” and “non-transitory computer-readable storage medium” should be construed to exclude only those types of transitory computer-readable media which were found in In re Nuijten to fall outside the scope of patentable subject matter under 35 U.S.C. § 101.

(176) The disclosure and claims do not describe only a particular outcome of mobile digital currency exchanges and an order flow process, but the disclosure and claims include specific rules for implementing the outcome of the mobile digital currency exchanges and the order flow process and that render information into a specific format that is then used and applied to create the desired results of the mobile digital currency exchanges and the order flow process, as set forth in *McRO, Inc. v. Bandai Namco Games America Inc*. (Fed. Cir. case number 15-1080, Sep. 13, 2016). In other words, the outcome of the mobile digital currency exchange can be performed by many different types of rules and combinations of rules, and this disclosure includes various embodiments with specific rules. While the absence of complete preemption may not guarantee that a claim is eligible, the disclosure does not sufficiently preempt the field of mobile digital currency exchanges at all. The disclosure acts to narrow, confine, and otherwise tie down the disclosure so as not to cover the general abstract idea of just a mobile digital currency exchange. Significantly, other systems and methods exist for mobile digital currency exchanges, so it would be inappropriate to assert that the claimed invention preempts the field or monopolizes the basic tools of mobile digital currency exchanges. In other words, the disclosure will not prevent others from providing mobile digital currency exchanges, because other systems are already performing the functionality in different ways than the claimed invention. Moreover, the claimed invention includes an inventive concept that may be found in the non-conventional and non-generic arrangement of known, conventional pieces, in conformance with *Bascom v. AT&T Mobility,* 2015-1763 (Fed. Cir. 2016). The disclosure and claims go way beyond any conventionality of any one of the systems in that the interaction and synergy of the systems leads to additional functionality that is not provided by any one of the systems operating independently. The disclosure and claims may also include the interaction between multiple different systems, so the disclosure cannot be considered an implementation of a generic computer, or to just “apply it” to an abstract process. The disclosure and claims may also be directed to improvements to software with a specific implementation of a solution to a problem in

the software arts.
(177) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean "one and only one" unless explicitly so stated, but rather "one or more." Moreover, where a phrase similar to 'at least one of A, B, and C' or 'at least one of A, B, or C' is used in the claims or specification, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Although the disclosure includes a method, it is contemplated that it may be embodied as computer program instructions on a tangible computer-readable carrier, such as a magnetic or optical memory or a magnetic or optical disk. All structural, chemical, and functional equivalents to the elements of the above-described various embodiments that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present disclosure, for it to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element is intended to invoke 35 U.S.C. § 112(f) unless the element is expressly recited using the phrase "means for" or "step for". As used herein, the terms "comprises," "comprising," or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

Claims

1. A method, comprising: receiving, by a computer-based system and through an authentication request, authentication data including an authentication token; transmitting, by the computer-based system, a verification link for mobile device verification as part of a device verification process; launching, by the computer-based system, a native application; determining, by the computer-based system using the native application, a connected state based on the authentication data and the authentication token; checking, by the computer-based system, for updates, in response to determining a connected state; entering, by the computer-based system, a forced update state, in response to determining an available update and an update now action; checking, by the computer-based system, for a server maintenance state; entering, by the computer-based system, a server maintenance state, in response to determining active server maintenance; determining, by the computer-based system, completed server maintenance; checking, by the computer-based system, for sign-in process state based on the sign-in data, in response to the determining the completed server maintenance; checking, by the computer-based system, for a signed-in state based on the sign-in data, in response to a sign in not being in process; checking, by the computer-based system, for prior state data of the native application, in response to the signed-in state being true; determining, by the computer-based system, whether the prior session is still live, in response to the prior state data existing; opening, by the computer-based system, a prior state based on the prior state data, in response to the prior session being still live, to minimize network congestion; displaying, by the computer-based system, a trading interface of the prior state comprising a portfolio drawer based on the authentication data, wherein the portfolio drawer is displayed

responsively to an interaction with each of a market view page, an accounts and settings page, and an instrument details page and comprises an instrument detail display and an account summary display; receiving, by the computer-based system and through the trading interface, an instrument detail selection; responsive to the receiving the instrument detail selection, generating, by the computer-based system, an advanced position interface to update the prior state; receiving, by the computer-based system and through the advanced position interface, a first order flow input for an asset of a trading account, the first order flow input disposed in a data entry field on the advanced position interface, the first order flow input including a first quantity associated with a currency pair, the currency pair including the asset; determining, by the computer-based system, a long direction maximum margin and a short direction maximum margin based on a current position value and an available margin; calculating, by the computer-based system, a zero position margin based on the long direction maximum margin and the short direction maximum margin; determining, by the computer-based system and through a dynamic margin position determination step, a first dynamic margin position associated with the trading account, wherein the dynamic margin position determination step comprises calculating the first dynamic margin position based on each of the current position value, the available margin of the trading account, and the first order flow input; displaying, by the computer-based system and through the advanced position interface, a dynamic margin position indicator including the zero position margin, the long direction maximum margin, the short direction maximum margin, and the first dynamic margin position, wherein the dynamic margin position indicator is scaled based on the long direction maximum margin and the short direction maximum margin; receiving, by the computer-based system and through the advanced position interface, a second order flow input including a second quantity associated with the currency pair, the second quantity different from the first quantity; responsive to the receiving the second order flow input, repeating the dynamic margin position determination step with the second order flow input to generate a second dynamic margin position, the second dynamic margin position based on a second position value determined from the currency pair, the second quantity, and the current position value; and adjusting, by the computer-based system and in real-time, the dynamic margin position indicator to transition the first dynamic margin position to the second dynamic margin position, the second dynamic margin position displayed on the dynamic margin position indicator relative to the current position value and between the short direction maximum margin and the long direction maximum margin, wherein the dynamic margin position indicator is dynamically displayed prior to the execution of the first order flow input or the second order flow input on a digital currency exchange, in response to a selection based on the dynamic margin position indicator, executing the second order flow input on the digital currency exchange.

2. The method of claim 1, wherein the first order flow input comprises at least one of a buy order, a sell order, an order modification, a leverage setting, or a close position command.

3. The method of claim 1, further comprising starting, by the computer-based system, an anonymous mode process in response to the sign-in data.

4. The method of claim 1, wherein the dynamic margin position indicator is adjusted from the first dynamic margin position to the second dynamic margin position prior to execution of an order reflective of the second order flow input.

5. The method of claim 1, wherein the second dynamic margin position is displayed on the dynamic margin position indicator relative to the current position value on a background bar of the dynamic margin position indicator in response to the adjusting the dynamic margin position indicator.

6. The method of claim 1, further comprising: receiving, by the computer-based system, a third order flow input including the currency pair and an adjusted quantity, and responsive to the receiving the third order flow input, adjusting, by the computer-based system and in real-time, the dynamic margin position indicator to transition the second dynamic margin position to a third

dynamic margin position, the third dynamic margin position displayed on the dynamic margin position indicator relative to the current position value and between the short direction maximum margin and the long direction maximum margin.

7. The method of claim 1, wherein the portfolio drawer is associated with a trading account.

8. The method of claim 1, further comprising: scaling a new position bar to the background bar; displaying the new position bar overlaid on the background bar based on the difference between the current position value and the new position value; and dynamically altering a length of at least one of the new position bar or the current position bar based on the order data, wherein the dynamic margin position indicator indicates the available margin that is assigned to the current position at least one of currently or will be after execution of the order data, and wherein the dynamic margin position indicator indicates the available margin that is returned to an available balance after executing the order data reducing the current position; and updating the display of the dynamic margin position indicator including at least one of the current position bar, the new position bar, the background bar or the arrow, in response to recalculating margin requirements or changing margin calculations.

9. The method of claim 1, wherein the real-time adjusting of the dynamic margin position indicator further comprises: calculating a size of a background bar based on a corresponding maximum margin value in a long direction and a corresponding maximum margin value in a short direction; displaying the background bar; calculating a zero position based on the long direction and the short direction; display a zero indicator at the zero position on the background bar; scaling the background bar to indicate the maximum margin available to an account holder in the long direction and the maximum margin available to the account holder in the short direction; displaying the current position bar based on the magnitude of the current position value relative to the corresponding maximum margin value; overlaying the current position bar on the background bar; receiving order data including a currency pair and a quantity; calculating a new position value based on the current position value and the quantity; displaying an arrow on the background bar indicating the new position value; and pointing the arrow toward the long direction or the short direction based on on an order type of the order data and the quantity.

10. The method of claim 1, further comprising receiving, by the computer-based system, sign-in data associated with the trading account.

11. The method of claim 1, wherein the trading account includes the asset and the available margin, the asset including the current position value based on a current value of the asset and a current quantity of the asset.

12. The method of claim 1, wherein the dynamic margin position indicator visualization improves understanding of the effect of the execution of the first order flow input or the second order flow input on a digital currency exchange.

13. A computer-based system comprising: one or more processors; and one or more tangible, non-transitory memories configured to communicate with the one or more processors, the one or more tangible, non-transitory memories having instructions stored thereon that, in response to execution by the one or more processors, cause the one or more processors to perform operations comprising: receiving, by the one or more processors and through an authentication request, authentication data including an authentication token; transmitting, by the one or more processors, a verification link for mobile device verification as part of a device verification process; launching, by the one or more processors, a native application; determining, by the one or more processors using the native application, a connected state based on the authentication data and the authentication token; checking, by the one or more processors, for updates, in response to determining a connected state; entering, by the one or more processors, a forced update state, in response to determining an available update and an update now action; checking, by the one or more processors, for a server maintenance state; entering, by the one or more processors, a server maintenance state, in response to determining active server maintenance; determining, by the one or more processors, completed

server maintenance; checking, by the one or more processors, for sign-in process state based on the sign-in data, in response to the determining the completed server maintenance; checking, by the one or more processors, for a signed-in state based on the sign-in data, in response to a sign in not being in process; checking, by the one or more processors, for prior state data of the native application, in response to the signed-in state being true; determining, by the one or more processors, whether the prior session is still live, in response to the prior state data existing; opening, by the one or more processors, a prior state based on the prior state data, in response to the prior session being still live, to minimize network congestion; displaying, by the one or more processors, a trading interface of the prior state comprising a portfolio drawer based on the authentication data, wherein the portfolio drawer is displayed responsively to an interaction with each of a market view page, an accounts and settings page, and an instrument details page and comprises an instrument detail display and an account summary display; receiving, by the one or more processors and through the trading interface, an instrument detail selection; responsive to the receiving the instrument detail selection, generating, by the one or more processors, an advanced position interface to update the prior state; receiving, by the one or more processors and through the advanced position interface, a first order flow input for an asset of a trading account, the first order flow input disposed in a data entry field on the advanced position interface, the first order flow input including a first quantity associated with a currency pair, the currency pair including the asset; determining, by the one or more processors, a long direction maximum margin and a short direction maximum margin based on a current position value and an available margin; calculating, by the one or more processors, a zero position margin based on the long direction maximum margin and the short direction maximum margin; determining, by the one or more processors and through a dynamic margin position determination step, a first dynamic margin position associated with the trading account, wherein the dynamic margin position determination step comprises calculating the first dynamic margin position based on each of the current position value, the available margin of the trading account, and the first order flow input; displaying, by the one or more processors and through the advanced position interface, a dynamic margin position indicator including the zero position margin, the long direction maximum margin, the short direction maximum margin, and the first dynamic margin position, wherein the dynamic margin position indicator is scaled based on the long direction maximum margin and the short direction maximum margin; receiving, by the one or more processors and through the advanced position interface, a second order flow input including a second quantity associated with the currency pair, the second quantity different from the first quantity; responsive to the receiving the second order flow input, repeating the dynamic margin position determination step with the second order flow input to generate a second dynamic margin position, the second dynamic margin position based on a second position value determined from the currency pair, the second quantity, and the current position value; and adjusting, by the one or more processors and in real-time, the dynamic margin position indicator to transition the first dynamic margin position to the second dynamic margin position, the second dynamic margin position displayed on the dynamic margin position indicator relative to the current position value and between the short direction maximum margin and the long direction maximum margin, wherein the dynamic margin position indicator is dynamically displayed prior to the execution of the first order flow input or the second order flow input on a digital currency exchange, in response to a selection based on the dynamic margin position indicator, executing the second order flow input on the digital currency exchange.

14. An article of manufacture including one or more non-transitory, tangible computer readable storage mediums having instructions stored thereon that, in response to execution by a computer-based system, cause the computer-based system to perform operations comprising: receiving, by the computer-based system and through an authentication request, authentication data including an authentication token; transmitting, by the computer-based system, a verification link for mobile device verification as part of a device verification process; launching, by the computer-based

system, a native application; determining, by the computer-based system using the native application, a connected state based on the authentication data and the authentication token; checking, by the computer-based system, for updates, in response to determining a connected state; entering, by the computer-based system, a forced update state, in response to determining an available update and an update now action; checking, by the computer-based system, for a server maintenance state; entering, by the computer-based system, a server maintenance state, in response to determining active server maintenance; determining, by the computer-based system, completed server maintenance; checking, by the computer-based system, for sign-in process state based on the sign-in data, in response to the determining the completed server maintenance; checking, by the computer-based system, for a signed-in state based on the sign-in data, in response to a sign in not being in process; checking, by the computer-based system, for prior state data of the native application, in response to the signed-in state being true; determining, by the computer-based system, whether the prior session is still live, in response to the prior state data existing; opening, by the computer-based system, a prior state based on the prior state data, in response to the prior session being still live, to minimize network congestion; displaying, by the computer-based system, a trading interface of the prior state comprising a portfolio drawer based on the authentication data, wherein the portfolio drawer is displayed responsively to an interaction with each of a market view page, an accounts and settings page, and an instrument details page and comprises an instrument detail display and an account summary display; receiving, by the computer-based system and through the trading interface, an instrument detail selection; responsive to the receiving the instrument detail selection, generating, by the computer-based system, an advanced position interface to update the prior state; receiving, by the computer-based system and through the advanced position interface, a first order flow input for an asset of a trading account, the first order flow input disposed in a data entry field on the advanced position interface, the first order flow input including a first quantity associated with a currency pair, the currency pair including the asset; determining, by the computer-based system, a long direction maximum margin and a short direction maximum margin based on a current position value and an available margin; calculating, by the computer-based system, a zero position margin based on the long direction maximum margin and the short direction maximum margin; determining, by the computer-based system and through a dynamic margin position determination step, a first dynamic margin position associated with the trading account, wherein the dynamic margin position determination step comprises calculating the first dynamic margin position based on each of the current position value, the available margin of the trading account, and the first order flow input; displaying, by the computer-based system and through the advanced position interface, a dynamic margin position indicator including the zero position margin, the long direction maximum margin, the short direction maximum margin, and the first dynamic margin position, wherein the dynamic margin position indicator is scaled based on the long direction maximum margin and the short direction maximum margin; receiving, by the computer-based system and through the advanced position interface, a second order flow input including a second quantity associated with the currency pair, the second quantity different from the first quantity; responsive to the receiving the second order flow input, repeating the dynamic margin position determination step with the second order flow input to generate a second dynamic margin position, the second dynamic margin position based on a second position value determined from the currency pair, the second quantity, and the current position value; and adjusting, by the computer-based system and in real-time, the dynamic margin position indicator to transition the first dynamic margin position to the second dynamic margin position, the second dynamic margin position displayed on the dynamic margin position indicator relative to the current position value and between the short direction maximum margin and the long direction maximum margin, wherein the dynamic margin position indicator is dynamically displayed prior to the execution of the first order flow input or the second order flow input on a digital currency exchange, in response to a selection based on the dynamic margin position indicator, executing the second order flow input on the digital currency exchange.